

Under Data Scarcity: Predictive Performance and Interpretability Stability Decouple for CKD Risk Prediction

**Predictive Performance (AUC)
stays comparatively stable**

0.776



0.748

N = 5,000

N = 250

**Interpretability Stability (Spearman)
declines sharply**

0.907



0.429

N = 5,000

N = 250

As sample size shrinks ($N = 5,000 \rightarrow N = 250$), model performance holds while interpretability method rankings become substantially less reproducible - adequate performance is not evidence of adequate interpretability.